

Assessment 2

Assessment 2

Machine Learning Classification and Model Comparison



Course Code: COSC2673 / COSC2793.

Weighting: This assessment is worth 40% of your final mark for this Course

Task type: Individual

Due date and time: Monday, 11 May 2026, 11:59 pm AEDT (Week 10)

Feedback: Feedback will be provided via Canvas

Click or tap on the relevant sections to learn more

▼ 1. Course Learning Outcomes Assessed

This assessment supports the following learning outcomes for COSC2673:

- CLO2: Justify the suitability of a range of machine learning methods for different kinds of problems.
- CLO3: Develop machine learning pipelines, including data preprocessing and feature engineering, for a range of applications contexts.
- CLO4: Rationalise major application areas of machine learning in contemporary professional and research contexts.
- CLO5: Articulate the ethical considerations and professional responsibilities involved in the application of machine learning.

This assessment supports the following learning outcomes for COSC2793:

- CLO2: Assess the suitability of a range of machine learning methods for different kinds of problems.
- CLO3: Design and implement machine learning pipelines, including data preprocessing and feature engineering, for a range of applications contexts.
- CLO4: Analogue major application areas of machine learning in contemporary professional and

research contexts.

- CLO5: Articulate and critique the ethical considerations and professional responsibilities involved in the application of machine learning.

▼ 2. Overview of Assessment

You will work with a real-world dataset to complete a machine learning task and practice the typical machine learning workflow, which includes:

- Defining the task and understanding the prediction target.
- Reading, splitting, and pre-processing a dataset.
- Conducting **exploratory data analysis (EDA)** to understand feature distributions and potential data issues (e.g., missing values).
- Implement three (3) machine learning models: **Decision Tree, Support Vector Machine (SVM), and Neural Network.**
- Design and apply a consistent evaluation framework (e.g, with hold-out validation **OR** k-fold cross-validation) for **model evaluation and hyperparameter selection.**
- Support your analysis with **quantitative results and visualisations.** This should include, but is not limited to:
 - performance curves (e.g., accuracy vs. hyperparameters),
 - and, for neural networks, training loss curves across epochs.
- Interpret the **behaviour of each model** based on experimental results. Your discussion should explain patterns such as underfitting, overfitting, convergence, or instability.
- **Evaluate** all models using consistent metrics (e.g., Accuracy and F1-score) and **compare** their performance systematically.
- **Select** a final model and **justify** your choice.
- Discuss any limitations and potential real-world implications.

To complete this assignment, you will need knowledge and skills from lectures and lab sessions from **Week 1 to 8**. You are encouraged to start with the sections covered so far in the lectures and labs, and progressively build on your work as you acquire new skills. Remember, developing a robust machine learning model takes time, so **start early**.

This assignment has **four deliverables**:

1. **Your Jupyter notebook**, including code, plots, results, and markdown explanations for the rationale for your approach, model design, critical analysis, and your final evaluation, along with clear instructions on how to run it.
2. **A PDF version of your Jupyter notebook**, identical to the .ipynb version.
3. **A CSV file of predictions for the test set**, generated by your chosen model in the format described below.
4. **A PDF report**, detailing your machine learning workflow and rationale, i.e., introduction, definition of the

task, description of the dataset, exploratory data analysis, splitting, pre-processing, machine learning techniques and model development (Decision Tree, SVM, Neural Network), hyperparameter tuning, model comparison, evaluation, justification of the selected model, and in-depth analysis, ethical considerations and professional responsibilities.

You may use GenAI tools for completing your assignment. If you do use a tool, you must reference the software, declaring the details of the prompts used, including a brief description of how it was used (e.g., debugging code, clarifying concepts, improving writing). Read here about how to [Acknowledge and Reference AI](#).

You are responsible for the correctness, validity, and integrity of all submitted work. Submitting content that is largely generated by GenAI without sufficient understanding or modification is **NOT** acceptable.

Plagiarism penalties apply if you did not do the work submitted. If the work you submit is highly similar to that of another student or any other source, because you both used GenAI tools or GenAI adapted from a source without proper attribution, this will also trigger plagiarism penalties, as it implies blind copying of GenAI output.

Our standard [Assessment Extensions and Late Policy](#) applies.

▼ 3. Assessment Task

Satellite-based wildfire monitoring systems allow the detection of fire events across different regions of the world. A key challenge is accurately determining the intensity level of each detected fire. If the intensity is underestimated, emergency response may be insufficient; if it is overestimated, resources may be wasted. Because fire intensity depends on many factors (e.g., weather conditions, environmental features, and location), machine learning can be used to classify fire intensity levels based on historical data.

In this assignment, you will develop a **machine learning model to predict the wildfire intensity levels across 7 regions and 35 countries (including Australia)**, based on environmental and observational features, including weather conditions, geographic information, and satellite-derived data.

The machine learning task we are interested in is:

“Predict the fire intensity level (4 classes) of a wildfire event based on the provided features.”

The dataset to develop your models is adapted from <https://www.kaggle.com/datasets/alitaqishah/wildfire-risk-dataset-2024-2025-7-regions> . Here are the training data [wildfire_cls_train_full.csv](#)  and test data [wildfire_cls_test_features.csv](#) .

Note that the training and test data cover the same 7 regions and 35 countries, which are representative of where the model will be applied. You **DO NOT** need to consider model generalization to regions or countries outside of these in this assessment.

The target column is "**fire intensity**", which represents the intensity level of a wildfire event. This is a categorical variable with four classes, which are encoded as follows:

0 = Low: Low-intensity fire events with minimal impact

1 = *Moderate : Medium-intensity fires with noticeable activity*

2 = *High: High-intensity fires with strong thermal signals*

3 = *Extremd: Very high-intensity fires with severe impact and risk.*

- Note that the target column is missing from the test data; you will be required to submit your predictions for the test data. You need to develop a model from the list of machine learning algorithms covered from Week 5 to Week 8 in the course (i.e., **Decision Tree, SVM, Neural Network**), where each element of the workflow is justified using **data analysis, hyperparameter analysis, performance discussion, and/or knowledge from relevant literature**.
- You are required to handle **missing values in training data** appropriately and justify your chosen method. You may use simple techniques covered in the labs and practicals.
- You are required to implement and compare **at least three (3) different machine learning models (Decision Tree, SVM, Neural Network)** and only use techniques taught in class up to Week 8 – inclusive. You should investigate how different models perform on this classification task and determine which model is most appropriate.
- For each of your models, investigate the effect of hyperparameters on your model, such as the effect of the "learning rate" parameter in Neural Network (set several candidate learning rates and run your model). You are required to discuss how to choose the best value for this hyperparameter, and its impact on the training and test performance of the model.
- **For [COSC2793 Computation Machine Learning] only:** For each model, investigate at least one additional hyperparameter and explain its impact on model behaviour and performance, linking experimental results to theoretical expectations (details are in the "Assessment Submission and Feedback");
- You are required to design an **evaluation framework** to assess model performance. This includes: (a) Splitting the training data into training and validation (e.g., with hold-out validation **OR** k-fold cross-validation); (b) Selecting appropriate evaluation metrics (e.g., Accuracy and F1-score).
- Finally, analyse the models and results using appropriate techniques and determine how adequate your model is for this task in the real world. Discuss any **limitations, ethical considerations, and professional responsibilities** if there are any.
- **Predict fire intensity level (4 classes: 0 = Low; 1 = Moderate; 2 = High; 3 = Extreme) of a wildfire event for the test set.**

You are only allowed to use techniques taught in class up to Week 8 (inclusive) for this assignment.

The dataset for this assignment is adapted from <https://www.kaggle.com/datasets/alitaqishah/wildfire-risk-dataset-2024-2025-7-regions> ↗, but was manually processed to clean the data, sample a smaller subset, and include selected features (along with some other steps to make this data unique from any dataset available online). The training data [wildfire_cls_train_full.csv](#) ↓ and test data [wildfire_cls_test_features.csv](#) ↓ contain the following features:

- latitude, longitude: Geographic coordinates of the fire event
- acq_date: Date of fire detection

- `acq_time`: Time of fire detection (in HHMM format)
- `year`, `month`
- `season`: Season of the year (e.g., summer, winter)
- `daynight`: Indicates whether the fire was detected during the day or night
- `region`: Geographic region of the fire event
- `country`: Country where the fire occurred
- `fire_type`: Type of fire event: Wildfire / Agriculture / Peatland / Savanna
- `satellite`: Satellite used for detection: NOAA-20 / Suomi-NPP
- `instrument`: Sensor/instrument used for detection: VIIRS / MODIS
- `brightness_k`: Brightness temperature (in Kelvin), related to fire activity
- `confidence`: Confidence level of the detection: high / nominal / low
- `temp_max_c`: Max air temperature on detection day (°C)
- `wind_max_kmh`: Max wind speed (km/h) — key for fire spread
- `precip_mm`: Precipitation — fire suppression proxy
- `humidity_pct`: Relative humidity (%)

The training dataset also includes the target variable:

- `fire_intensity`: The intensity level of the wildfire (categorical variable with four (4) classes: **0 = Low; 1 = Moderate; 2 = High; 3 = Extreme**)

The original dataset was under a [CC0: Public Domain](#)  license. However, the dataset herein can only be used for the purpose of this assignment. Sharing or distributing this data or using this data for any other commercial or non-commercial purposes is prohibited.

▼ 4. Timeline

To help you prepare for the assessment, please consider the following activities:

Timeline	Activities
Week 7	• Review specifications and Download datasets. • Explore the dataset and perform initial exploratory data analysis (EDA) • Perform data preprocessing (e.g., handling missing values, basic feature processing).
Week 8	• Develop a baseline machine learning pipeline (e.g., Decision Tree).

Timeline	Activities
	• Evaluate initial model performance. • Implement and compare multiple models (e.g., SVM, Neural Network). • Conduct hyperparameter tuning for selected model(s).
Week 9 to Monday of Week 10 (Submission Date: 11 May 2026)	• Analyse and compare model performance. • Finalise the machine learning pipeline. • Generate predictions on the test dataset. • Complete the final report. • Submit!

▼ 5. Assessment Submission and Feedback

You should submit two files: (1) a single zip file containing the following and (2) a PDF of the report (i.e. the report is submitted twice, separate and within the zip file):

1. **Your Jupyter notebook**, including code, plots, results, and markdown explanations for the rationale for your approach, critical analysis, and your final evaluation, along with clear instructions on how to run it.
2. **A PDF version of your Jupyter notebook**, identical to the .ipynb version.
3. **A CSV file of predictions for the test set**, generated by your chosen model (only one model). The file should be named '{student_number}_predictions.csv' containing a single column named **'fire_intensity'** (note the use of an underscore _ between *fire* and *intensity*), ensuring that the predictions are in the same order as the samples in 'wildfire_cls_adapt_test_nolabel.csv'. You may use code similar to:

```

y_pred = model.predict(X_test)
df_y_pred = pd.DataFrame({'fire_intensity': y_pred})
df_y_pred.to_csv('s1234567_predictions.csv', header=True, index=False)

```
4. **A PDF report**, detailing your machine learning workflow, in-depth model analysis, and a detailed written report. The report is expected to cover following: (O) Introduction; (A) Task Definition and Dataset Description; (B) Exploratory Data Analysis (EDA) and Data Handling; (C) Model Development and Performance Analysis (Quantitative Analysis and Visualization); (D) Model Comparison; (E) Final Model Selection, Justification, and Evaluation; (F) Ethical considerations and professional responsibilities (if applicable); (G) Conclusion; (H) References.
 - The report should be at **minimum 8 pages and at most 16 pages** (including all figures, tables, references, and appendices, etc).

- **For [COSC2793 Computation Machine Learning]**, the report should be at **minimum 10 pages and at most 20 pages** (including all figures, tables, references, and appendices, etc).
5. The report should address the previous list, but also the following questions:

A. Task and Dataset Understanding

- What is the task and prediction target?
- What are the key characteristics of the dataset?
- What challenges might this dataset present (e.g., missing values)?

B. EDA and Data Handling

- How are the features correlated with each other? How are the features correlated with the target? Are all features necessary or important?
- What are the preprocessing steps? Describe how the data is prepared for modelling, including handling missing values, encoding categorical variables, and scaling numerical features (if applicable).

C. Model Development and Performance Analysis (Quantitative Analysis and Visualisation)

- For each model (**Decision Tree, SVM, Neural Network**), you must: (a) Explain why Decision Tree, SVM, and Neural Network were selected (Pros and Cons for each model, and why it is appropriate (or not) for this specific dataset); (b) Conduct controlled experiments by varying key hyperparameters; (c) Present quantitative results using visualisations, e.g, for Neural Networks, training loss curves across epochs; (d) Provide detailed analysis of model behaviour based on experimental results, e.g., evidence of underfitting or overfitting, sensitivity to hyperparameters, etc.
- Your analysis must address the following model-specific questions:

(a) Decision Tree: (1) How does model complexity (e.g., `max\_depth`) affect performance? (2) Visualization the obtained tree structure; (3) What insights can be drawn from the tree structure? **(4) For [COSC2793 Computation Machine Learning] only:** How would post-pruning (or limiting depth) affect model performance?

(b) SVM: (1) How does performance differ between "linear" and "RBF" kernels? (2) How does the regularization parameter "C" affect training vs validation performance? **(3) For [COSC2793 Computation Machine Learning] only:** How does any other hyperparameter affect the decision boundary?

(c) Neural Network: (1) How does the "learning rate" affect model training behaviour? (2) Which learning rate leads to stable convergence? (3) What do the loss curves reveal about the optimisation process? **(4) For [COSC2793 Computation Machine Learning] only:** How does any other hyperparameter affect the model performance?

D. Model Comparison

- Which model performs best in the developed evaluation framework?

- How do the models differ in how they separate the classes? For example, which models produce simpler vs. more complex boundaries, and which are more sensitive to hyperparameters or more stable based on your results?
- Do more complex models always perform better than simpler ones in this task? Why or why not?
- **For [COSC2793 Computation Machine Learning] only:** Do your experimental results match what you would expect based on theory? For instance, a more complex model may fit training data better, but a simpler model may generalise better, consistent with theory.

E. Final Model Selection, Justification, and Evaluation

- Which model is selected as the final model, and how to obtain the test results?
- Why is this model preferred, beyond just performance?

▼ 6. Assessment Criteria

Provide the assessment criteria here. This may include a rubric or specific guidelines on how the assessment will be graded.

Component	Criteria	Weighting
Task and Dataset Understanding	Assesses how clearly the student defines the machine learning task, identifies the prediction target, and demonstrates understanding of key dataset characteristics and challenges.	5%
EDA and Data Handling	Assesses the quality of exploratory data analysis and appropriateness of data handling, including feature relationships, preprocessing, and data splitting.	5%
Model Development and In-depth Analysis	Assesses model development, experimental design, hyperparameter tuning, model evaluation, visualisations, and depth of analysis.	40%

Component	Criteria	Weighting
Model Comparison, Selection, and Justification	Assesses the ability to compare models and justify the final model choice.	10%
Prediction of Test Data	Assesses the correctness, reproducibility, and formatting of predictions submitted for the test dataset, including adherence to the required file structure.	10%
Report Quality	Assesses the clarity, organisation, and professionalism of the report, including completeness of the machine learning workflow, depth and quality of analysis, detailed discussion, and effectiveness of communication.	30%

► 7. Academic Integrity

For more information, please consult the course guide or contact your course coordinator.

Criteria	Ratings					Pts
Task and Dataset Understanding Assesses how clearly the student defines the machine learning task, identifies the prediction target, and demonstrates understanding of key dataset characteristics and potential challenges.	5 to >4.0 Pts High Distinction The task and prediction target are clearly and precisely defined. The student demonstrates a strong understanding of dataset characteristics, including feature types and structure. Potential challenges (e.g., missing values) are correctly identified and briefly	4 to >3.0 Pts Distinction The task and prediction target are correctly defined. The dataset is described with reasonable understanding of its main characteristics with minor ambiguity. Some challenges are identified but explanation is limited.	3 to >2.0 Pts Credit The task is described in a basic way but lacks clarity or precision. Dataset characteristics are mentioned but without meaningful interpretation. Challenges are only briefly noted or not clearly	2 to >0.0 Pts Pass The task definition is incomplete or unclear. Dataset description is minimal and mostly descriptive. Little or no understanding of challenges is shown.	0 Pts NN The task is incorrectly defined or missing. Dataset description is absent or incorrect. No evidence of understanding is demonstrated.	5 pts
EDA and Data Handling Assesses the quality of exploratory data analysis and appropriateness of data handling, including feature relationships, preprocessing, and data splitting.	5 to >4.0 Pts High Distinction EDA clearly identifies relationships between features and the target using appropriate methods. Data handling steps (missing values, encoding, scaling) are appropriate and well-justified. Data splitting is correctly implemented	4 to >3.0 Pts Distinction EDA is generally correct and identifies some useful relationships. Data preprocessing is mostly appropriate but justification may be limited. Data splitting is correct but not deeply explained.	3 to >2.0 Pts Credit EDA is basic and mostly descriptive, with limited insight into relationships. Preprocessing steps are applied but lack clear reasoning. Data splitting is mentioned but not clearly justified.	2 to >0.0 Pts Pass EDA is minimal or superficial. Data handling steps are unclear or partially incorrect. Data splitting is poorly explained or incorrectly implemented.	0 Pts NN EDA is missing or incorrect. Data handling is inappropriate or absent. No clear data preparation process is shown.	5 pts

Criteria	Ratings					Pts
	and clearly explained.					
Model Development and In-depth Analysis Assesses model development, experimental design, hyperparameter tuning, visualisations, and depth of analysis.	40 to >28.0 Pts High Distinction All three models (Decision Tree, SVM, Neural Network) are correctly implemented with clear justification of their selection. Hyperparameter experiments are systematic and well-designed, with proper use of validation. Visualisation and quantitative analysis are complete, and the analysis clearly explains model behaviour.	28 to >20.0 Pts Distinction Models are implemented correctly with reasonable justification. Hyperparameter tuning is performed, but may not be fully systematic. Visualisations are mostly complete, and quantitative analysis shows good understanding with some depth.	20 to >12.0 Pts Credit Models are implemented but justification is limited. Some hyperparameter tuning is performed, but experiments lack structure. Visualisations are incomplete or weakly explained, and analysis is mostly descriptive.	12 to >0.0 Pts Pass Model implementation is partial or inconsistent. Hyperparameter tuning is minimal or unclear. Visualisations are missing or poorly interpreted, and analysis lacks depth.	0 Pts NN Models are incorrectly implemented or missing. No meaningful experiments or visualisations are provided. No valid analysis is present.	40 pts

Criteria	Ratings					Pts
Model Comparison, Selection and Justification Assesses the ability to compare models through appropriate evaluation, analyse differences in model behaviour, and justify the final model choice based on multiple factors beyond performance.	10 to >8.0 Pts High Distinction All three (3) models (Decision Tree, SVM, Neural Network) are compared comprehensively using validation results with clear reference to appropriate metrics. The analysis considers multiple aspects, including sensitivity to hyperparameters. Differences in model behaviour are clearly explained and supported by evidence. Final model selection is well-justified using multiple factors (e.g., performance, stability, complexity). Where applicable, results are linked	8 to >6.0 Pts Distinction Models are compared clearly using appropriate evaluation metrics. Some discussion of performance differences and model behaviour is provided, including basic considerations. Final model selection is reasonable and supported by relevant results, but justification may rely mainly on performance with limited deeper analysis.	6 to >4.0 Pts Credit Comparison is present but primarily based on performance metrics only. Limited discussion is provided on differences in model behaviour and stability. Final model selection is stated, but justification is minimal or lacks supporting reasoning.	4 to >0.0 Pts Pass Comparison between models is weak, unclear, or incomplete. Analysis is largely descriptive with little interpretation of results. Final model selection lacks clear reasoning and shows limited understanding of model differences.	0 Pts NN No meaningful comparison is provided. Models are not evaluated properly or results are missing. Final model selection is absent or completely unjustified, with no supporting analysis.	10 pts

Criteria	Ratings					Pts
	to theoretical expectations, and any inconsistencies are explained.					
Prediction of Test Data Assesses the correctness, reproducibility, and formatting of predictions submitted for the test dataset, including adherence to required file structure.	10 to >8.0 Pts High Distinction Predictions are correctly generated and fully reproducible. The CSV file strictly follows all required specifications, including correct file format, correct number of rows, correct column name (fire_intensity), and correct ordering of samples. The prediction performance ranks in the top ~60% of the class (approximately).	8 to >6.0 Pts Distinction Predictions are correctly generated with only minor formatting or reproducibility issues (e.g., small inconsistencies that do not affect usability). The prediction performance ranks in the bottom ~40 % of the class (approximately).	6 to >4.0 Pts Credit Predictions are generally correct and complete, with some issues in formatting, consistency, or reproducibility.	4 to >0.0 Pts Pass Predictions are partially correct or incomplete, with noticeable errors in format, structure, or process.	0 Pts NN Predictions are incorrect, missing, or unusable.	10 pts
Report Quality Assesses the clarity, organisation, and professionalism of the report, including completeness of the machine learning	30 to >24.0 Pts High Distinction Report is highly accurate, comprehensive, and well structured. All sections of the machine learning workflow are clearly	24 to >18.0 Pts Distinction Report is accurate and well organised. Most sections are clearly presented, with good explanation and analysis. Justification of decisions is	18 to >12.0 Pts Credit Report communicates the work adequately. Key sections are included but may lack detail, clarity, or depth. Analysis is present but limited, and	12 to >0.0 Pts Pass Report is understandable but limited in structure, clarity, or completeness. Important parts of the workflow are missing or poorly explained.	0 Pts NN Report is poorly written, incomplete, or difficult to follow. Major components are missing, and communication quality is poor.	30 pts

Criteria	Ratings					Pts
workflow, depth and quality of analysis, detailed discussion, and effectiveness of communication.	presented, including task definition, dataset description, EDA, preprocessing, model development, evaluation, and analysis. Writing is concise, clear, and insightful. Strong justification is provided for all decisions, and analysis demonstrates deep understanding. Communication is professional, and references are appropriate and consistently formatted.	generally clear, with some insight into model performance. Minor issues in clarity, depth, or conciseness may be present.	justification of decisions may be incomplete or superficial.	Analysis is weak or minimal.		
Total points: 100						